



Zhou Liu

Ph.D. Student in Data Science and Engineering

Peking University | Beijing | 134-7613-5561 | zhouliu.919@qq.com

[Homepage](#) | [GitHub: DeepThinkingZhouLiu](#) | [Google Scholar](#)

Education

Peking University, Academy for Advanced Interdisciplinary Studies

Ph.D. in Data Science and Engineering · Advisor: Prof. Wentao Zhang

Beijing
Sep. 2025 – Expected Jun. 2029

South-Central Minzu University, School of Computer Science

M.S. in Computer Science and Technology

Wuhan
Sep. 2022 – Jun. 2025

South-Central Minzu University, School of Computer Science

B.S. in Software Engineering

Wuhan
Sep. 2018 – Jun. 2022

Research Interests

Large language models and post-training; reinforcement learning, heuristic learning (HL), and alignment; agent planning, tool use, multi-agent collaboration, and agentic RL; data analytics and data governance agents; VLM agents, computer user agents, and multimodal interaction; multimodal LLMs and cross-modal understanding; benchmark construction, synthetic data, and automated evaluation.

Skills

LLMs and Post-training: Proficient with PyTorch and Transformers; familiar with SFT, instruction tuning, LoRA, DPO, RLHF, and end-to-end post-training workflows; familiar with Qwen and LLaMA families; experienced with distributed training using DeepSpeed and ZeRO.

Reinforcement Learning and Alignment: Familiar with PPO- and GRPO-style optimization, preference learning, reward design, semantic rewards, and alignment evaluation; research experience in applying reinforcement learning to language model capability expansion and agent optimization.

Agents: Familiar with ReAct, plan-and-execute, tool use, workflow orchestration, and multi-agent collaboration; experienced in dynamic task decomposition, tool registration, result verification, and closed-loop feedback systems; familiar with Camel and Minion-Agent.

Engineering and Evaluation: Experienced in benchmark construction, synthetic data generation, automatic evaluation, hybrid retrieval, and deep retrieval; built experiment systems for complex data analytics pipelines, reproducibility, and result analysis.

Internships

ByteDance ByteDance, Volcano Engine, Data Model Team

Algorithm Intern

Beijing
Jul. 2025 – Present

- Worked on data analytics agents and data governance scenarios, including benchmark construction, dataset development, and evaluation pipelines that supported ACL, COLM, and KDD projects.
- Explored structured, unstructured, and cross-modal data analysis tasks with multi-agent collaboration, tool calling, and retrieval-augmented strategies.
- Contributed to data processing pipelines and experimental systems that improved benchmark generation, reproducibility, and evaluation efficiency.

ByteDance ByteDance, Douyin Intelligent Creation Team

Graphics and Vision Algorithm Intern

Beijing
May 2024 – Dec. 2024

- Built inference pipelines for AIGC image-generation and editing models in Douyin and Qingyan, covering model integration, pre/post-processing, operator adaptation, numerical alignment, and visual-quality validation for fast model iteration.
- Profiled and optimized production inference across end-to-end latency, throughput, peak GPU/CPU memory, model loading, and redundant computation; improved efficiency and stability through batching, cache reuse, and tiered fallback strategies.
- Designed a unified multi-model inference protocol and workflow orchestration framework, standardizing model, operator, parameter, and resource loading while supporting heterogeneous model composition, failure fallback, and reuse across applications.

Selected Projects

RSI / HarnessEvoGym

[GitHub](#)

Trusted and reproducible recursive self-improvement platform for agent harnesses

TypeScript / Coding Agents / Docker

- Designed a decoupled Target $\times$ Environment $\times$ EvolutionRecipe architecture, with a frozen Controller enforcing mutation leases, diff guards, isolated evaluation, strict promotion, rollback, and lineage tracking.
- Implemented five population topologies and modular search strategies, with adapters for MSA Cowork/Reasoning, Codex CLI/DeepSeek Harness, and OmegaUse-OfficeVal; added checkpoint resume and sealed-final evaluation.

OpenDCAI / Paper2Any

[GitHub 2.1k stars](#) · [Live Demo](#)

Multimodal scientific content generation with editable research artifacts

Python / TypeScript

- Contributed to a multimodal workflow system that takes paper PDFs, screenshots, or text as input and generates architecture diagrams, technical roadmaps, experimental figures, and slide decks.
- Supported Paper2Figure, Paper2PPT, Paper2Poster, and Paper2Citation with editable DrawIO and PPT outputs through knowledge-base retrieval, document parsing, and multi-stage generation pipelines.

Paper Agent (AutoResearch)

Research in Progress

Research-agent prototype for paper understanding, research planning, and experiment loops

Python / LLM / VLM

- Designing an AutoResearch workflow intended to connect paper parsing, citation graphs, related-work organization, problem discovery, hypothesis formation, and experiment-plan decomposition into a traceable research loop.
- Currently exploring LLM/VLM reasoning, tool use, structured notes, and self-reflection evaluation, with planned validation against retrieval evidence, code executability, and experimental feedback.

OpenDCAI / DistillWheel

Core Developer · Private Repository

Vertical-task data distillation and training-data flywheel framework

Python / LLM / Agents

- Architected a configuration-driven distillation framework, using RunPlan to orchestrate TeacherHub, EnvHub, StoreHub, normalizers, quality gates, and artifact builders from seed data and model/environment trajectories to verified training samples.
- Implemented math, BrowserGym, and deep-research pipelines (OpenThoughts, Big-Math, Numina-TIR, QUEST, OpenResearcher, InfoSeek, and WebSailor), with tool execution, trajectory logging, automated verification, and SFT/RL/agent artifact generation.

OpenDCAI / DataFlow

[GitHub 3.2k stars](#)

Data processing, retrieval augmentation, and data preparation pipeline

Python

- Contributed to DataFlow Agent engineering and maintenance, including console fixes, pipeline.compile() preflight checks, and debugging-efficiency improvements ([representative PR #214](#)).
- Developed private-domain structured data synthesis, adaptive data cleaning, data quality assessment, and a hybrid retrieval pipeline based on DeepSearch.

Publications

[FIRST]	Source-Grounded Semantic Reinforcement Learning for Low-Resource Target-Language Generation	EMNLP 2026 Main · CCF-B
[FIRST]	DataCross: A Unified Benchmark and Agent Framework for Cross-Modal Heterogeneous Data Analysis	EMNLP 2026 Findings · CCF-B
[FIRST]	UniDataBench: Evaluating Data Analytics Agents Across Structured and Unstructured Data	ACL/26 Main Oral · Top 5% · CCF-A
[FIRST]	Paper2SysArch: Structure-Constrained System Architecture Generation from Scientific Papers	CVPR 2026 · CCF-A
[FIRST]	InsightBenchMaker: Towards Generating Evolving and High-Fidelity Benchmarks for Data-Analysis Agents	ACL 2026 Findings · CCF-A
[FIRST]	SciFlow-Bench: Evaluating Structure-Aware Scientific Diagram Generation via Inverse Parsing	ACL 2026 Main · CCF-A
[FIRST]	Reinforcement Learning with Semantic Rewards Enables Low-Resource Language Expansion without Alignment Tax	ACL 2026 Findings · CCF-A
[FIRST]	DataFlow: An LLM-Driven Framework for Unified Data Preparation and Workflow Automation in the Era of Data-Centric AI	SIGMOD 2027 · CCF-A

Honors and Awards

- 2026: Award, CVPR Vision-Language Agent Security Challenge
- 2023: National Second Prize, Huawei Cup China Graduate Mathematical Contest in Modeling
- 2023: National Third Prize and Provincial Third Prize, Teddy Cup Data Mining Challenge
- 2023: First-Class Graduate Academic Scholarship
- 2022: Provincial Third Prize, China Internet+ Innovation and Entrepreneurship Competition
- 2022: Provincial Second Prize, Chinese College Students Mathematics Competition
- 2022: Provincial Third Prize, Lanqiao Cup Algorithm Competition